3D Image Reconstruction in Microwave Tomography Using an Efficient FDTD Model

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***Abstract*—This letter describes an efficient three-dimensional (3D) image reconstruction algorithm for microwave tomography. Using the thin-wire approximation with resistive voltage sources (RVS), a realistic FDTD model is created for the transmitting and receiving antennas. To verify the algorithm, image reconstruction is made from experimental data using a cylindrical test object made of sunflower oil. In the results, successful image reconstruction has been shown, indicating that the FDTD model is applicable. For comparison, a reconstruction with a two-dimensional (2D) version of the algorithm was made. A significant increase in accuracy of the reconstructed object was seen for the 3D version.**

***Index Terms*—Antenna modeling, finite-difference time-domain (FDTD) methods, microwave imaging, microwave measurements, tomography.**

I. Introduction

In recent years, microwave imaging has attracted increasing interest within the research community. A number of potentially useful applications have been identified in areas as diverse as biomedical imaging, ground surface mapping of the Earth, detection of underground objects, nondestructive testing of materials, the detection of defects and cracks in construction materials, etc. Development has been accelerated with the increase in available computational power and algorithm improvements during the last decades. For objects with low contrast in the dielectric parameters, it is possible to use fast reconstruction algorithms utilizing, for example, the Born approximation. For high-contrast objects, the measured electrical fields on the receiving antennas are nonlinearly related to the dielectric properties in the reconstruction region. Consequently, it is necessary to resort to more computationally demanding iterative algorithms to maintain the reconstruction accuracy, [1]. These algorithms usually compare the measured data against a numerical simulation of the system, and the dielectric profile is iteratively updated based on the difference between the simulated and the measured data. Even though this comparison requires a realistic numerical model for the best accuracy, most of the published work has used two-dimensional (2D) models with different calibration procedures of the data. Largely, this can be attributed to a significant increase in the computational load when moving from 2D to three-dimensional (3D) modeling. Some recent 3D work can, however, be found in the literature, [2]–[5]. Furthermore, the reconstruction problem is inherently ill-posed, and regularization is usually required. The regularization enforces physical bounds on the reconstructed solution in order to reduce the number of possible solutions, and therefore to mollify the ill-posedness of the inverse problem. A common technique for regularization is to enforce a certain smoothness on the reconstructed image with the Tikhonov method.

In this paper, we introduce and evaluate an efficient 3D reconstruction algorithm based on realistic finite-difference time-domain (FDTD) modeling of the antenna array with the thin-wire approximation and the resistive voltage source (RVS) to model both transmitting and receiving antennas. Nonactive antennas were also included in order to account for the mutual coupling. A cylindrical object made of sunflower oil was used as a test object in the 3D reconstruction procedure. For comparison, the same test object was used with a 2D version of our reconstruction algorithm.

This paper builds on our earlier work on microwave tomography. Using a 2D algorithm similar to the one described in this paper, results from our experimental imaging prototype were published in [6]. The reconstruction accuracy in relation to the spectral content and the object size was investigated in [7]. In 2D, we also explored the use of *a priori* data in the reconstruction process [8], and deduced a scaling factor for the gradients for improved convergence in lossy media [9]. In [10], we initiated the work on 3D tomographic system by validating our forward 3D model with the experimental data obtained from our prototype apparatus.

This letter is organized as follows. Section II describes the theoretical background of our work, FDTD methods together with the various updating equations for the minimization procedure. In Section III, the results and discussions are presented. Here, we also make a comparison of the 3D and 2D results. The conclusions are presented in Section IV.

II. Theoretical Background

In this work, an iterative electromagnetic time-domain inversion algorithm has been used for reconstructing the dielectric parameters of the target object under test. It is based on FDTD simulations of the electromagnetic problem and computation of the adjoint Maxwell problem to compute gradients used in an optimization algorithm. The adjoint field is used extensively in different types of inverse problems [11], [12]. The algorithm used in this paper has been described in detail elsewhere for the 2D case [1], [7]. Slightly different variations of the algorithm are also described by other authors [2], [13]–[15] where reconstruction of the magnetic permeability is also considered. In [2],

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[14], and [15], the full-wave problem is taken into account, but only with 2D examples in [14] and [15]. It is relatively straightforward to extend the 2D algorithm to 3D, and a brief overview of our algorithm together with the differences between the 3D and 2D versions are presented here. The major advantage with the 3D configuration is that it allows for a more realistic modeling of the antennas and the geometry.

Our basic idea is to use scattering measurements of ultrawideband transient pulses for several transmitter/receiver combinations surrounding the target region and to compare the measured data with a numerical simulation of the system. The difference between the measured and the simulated signals is used to update the dielectric properties in the target region. As the reconstruction of the target object is iteratively refined, the simulated and measured signals converge. The assumption is that when the difference is sufficiently small, the reconstruction is completed. In other words, the aim of the reconstruction procedure is to minimize the objective functional F defined as

$$F(\boldsymbol{\epsilon},\boldsymbol{\sigma}) = \int_0^T \sum_{m=1}^{M} \sum_{n=1}^{N} |\mathbf{E}_{mn}(\epsilon,\sigma,t) - \mathbf{E}_{mn}^{\mathrm{m}}(t)|^2 dt \qquad (1)$$

where $\mathbf{E}_{mn}(\epsilon,\sigma,t)$ is the calculated field from the computational model and $\mathbf{E}_{mn}^{\mathrm{m}}(t)$ is the corresponding measured data when antenna number m has been used as transmitter and antenna n as receiver. M is the number of transmitters, and N is the number of receivers. In a 2D TM-mode formulation, only one spatial field component is used in (1), but for the 3D formulation, it is necessary to include all three spatial components.

In search for minimum, the functional is differentiated with respect to the dielectric components by a first-order perturbation analysis. The Fréchet derivative of the functional is used to define the gradients in every grid point inside the reconstruction domain as

$$G_\epsilon = 2\sum_{m=1}^{M}\int_0^T \tilde{\mathbf{E}}_m \cdot \partial_t \mathbf{E}_m dt \qquad (2)$$

$$G_{\sigma/\langle\sigma\rangle} = 2\,\langle\sigma\rangle \sum_{m=1}^{M}\int_0^T \tilde{\mathbf{E}}_m \cdot \mathbf{E}_m dt \qquad (3)$$

with the transmitting antenna m, $\mathbf{E}_m$ is the numerically simulated E-field in the reconstruction domain for the simulation time T, and $\tilde{\mathbf{E}}_m$ is the solution to the adjoint Maxwell's equations with the residual between the computed and the measured fields as sources. In the 3D algorithm, the product in the gradient is interpreted as a scalar product, reducing to a normal multiplication in the TM-mode 2D case. The parameter $\langle\sigma\rangle$ is a prescaling parameter necessary when imaging the conductivity and the permittivity simultaneously. Different units of the permittivity and conductivity introduce an imbalance between the gradients. This has to be compensated for, and the prescaling is given by [13]

$$\langle\sigma\rangle = \left(\int_0^\infty \sum_{m=1}^{M}\sum_{n=1}^{N} |\hat{\mathbf{E}}_{mn}(\omega)|^2 d\omega\right)^{-1} \times \int_0^\infty \sum_{m=1}^{M}\sum_{n=1}^{N} |\hat{\mathbf{E}}_{mn}(\omega)|^2 \omega d\omega. \qquad (4)$$

Here, $\hat{\mathbf{E}}_{mn}$ denotes the Fourier transformation of the measured time-domain electrical field from transmitter m on receiver n. In [9], we discuss the prescaling strategy more systematically and also extend it to improve convergence in lossy media by introducing a radial prescaling of the gradient. In [16], they took a different approach in order to determine the gradient prescaling by making a parametric study of the error measures of the reconstruction versus the scaling parameter. Finally, the gradients in (2) and (3) are used with the conjugate-gradient method together with the successive parabolic interpolation line search to find the optimal step length. The reconstruction procedure is then iterated with the objective functional used as a measure to monitor the convergence and to determine when to quit the algorithm.

The numerical simulations are made with the FDTD algorithm, [17]. In our previous work and in the example below with the 2D algorithm, the transmitters were modeled using a hard source model, where the field strength was simply specified at the desired position. At the receiver locations, the field values were sampled directly. Due to its simplicity, the hard source model is probably the most frequently used point source model in FDTD. By contrast, the 3D algorithm allows more realistic modeling of the antennas. In this work, we used the thin-wire approximation to model the monopoles [18] and RVS with 50 Ω impedance to model the feed at the transmitting, receiving, and inactive antennas [19], [20]. Furthermore, the ground plane was modeled as a perfect electric conductor. The walls of the tank have dielectric properties close to those of air and were therefore neglected in the numerical model. Instead, the FDTD domain was truncated by the convolutional perfectly matched layer (CPML) absorbing boundary condition so that the region outside the antenna array was treated as empty space. A more detailed discussion and validation of the FDTD antenna array model with a comparison against experimental data can be found in our previous work [10].

As already mentioned, the solution of the inverse problem relies on the comparison between the measured and the simulated scattering data. Since the numerical model is never perfect, a calibration procedure of the measured data was performed such that

$$E_{\mathrm{cal}}^{\mathrm{m}}(f) = \frac{S_{\mathrm{scat}}^{\mathrm{m}}(f)}{S_{\mathrm{ref}}^{\mathrm{m}}(f)}\, E_{\mathrm{ref}}^{\mathrm{s}}(f). \qquad (5)$$

Here, $S_{\mathrm{scat}}^{\mathrm{m}}$ is the measured reflection/transmission coefficients of the test object, $S_{\mathrm{ref}}^{\mathrm{m}}(f)$ is a reference measurement of an empty system, and $E_{\mathrm{ref}}^{\mathrm{s}}(f)$ is the corresponding simulated reference signal. $E_{\mathrm{cal}}^{\mathrm{m}}(f)$ is used for comparing with the FDTD simulations in the reconstruction process [1].

In our experimental prototype, the antennas are positioned in a plane. The possibility to accurately reconstruct out-of-plane objects is very limited: To do so, it would be necessary also to make additional measurements outside the antenna plane. To allow imaging with the 3D algorithm of a test object with finite height, we implemented a heuristic pseudo-3D technique that assumed constant properties of the test object as a function of height z above the ground plane. The gradient computed in the grid cell plane immediately above the ground plane were copied upward to the height of the test object. This method needs *a priori* information about the height of the reconstructed target. Since the reconstruction problem is both nonlinear and ill-posed, the resulting image strongly depends on the adopted regularization technique, the initialization of the reconstruction and also the spectral content of the pulse [7]. In the following examples, we started the reconstruction procedure from the background dielectric values of empty space, i.e., $\epsilon_r = 1.0$ and $\sigma = 0.0$ S/m. We also used the spectral content that produced the best reconstructed image. In the vicinity of the antennas, artifacts are often introduced. We therefore made the radius of the reconstruction domain smaller than the antenna array and also attenuated the gradients close to the boundary of the domain by multiplying the gradient with the function

$$1 - \exp\left[-0.15(R_{rd} - r)^2\right]. \tag{6}$$

Here, r is the radial distance from the center of the reconstruction domain with radius R_{rd}.

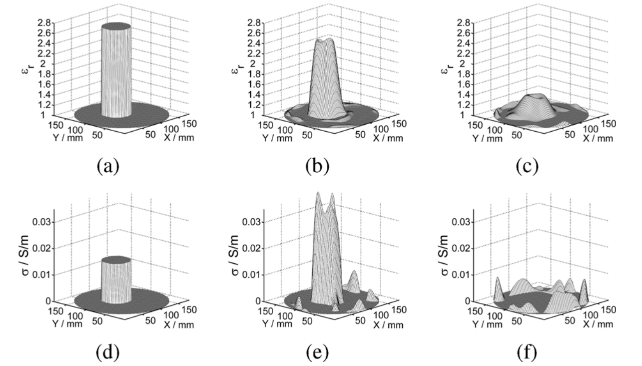


Fig. 1. (a) A cross section of the relative permittivity of the original target. (b) The reconstruction of the relative permittivity with the 3D algorithm and (c) with the 2D version. (d) A cross-sectional slice of the original conductivity and (e) a corresponding reconstruction with the 3D algorithm and (f) the 2D algorithm.

III. Results and Discussion

A. Experimental Arrangement

The experimental system consists of 20 monopoles each of length 19.5 mm and diameter 0.8 mm arranged in a circle of 200 mm diameter on a finite ground plane with size 250 mm × 250 mm. This configuration results in a resonance frequency of 3.5 GHz for each monopole. Furthermore, the assembly is enclosed in a container (370 mm^3) made of 1-cm-thick perspex sheets. The tank makes it possible to have both the antennas and the target object immersed in a matching liquid. This is important when investigating objects with a high contrast in dielectric properties to air, where strong reflections from the surface will reduce the penetration of the electric fields into the target and therefore reduce the ability to reconstruct the interior of the object. For example a matching liquid is required to give a better impedance matching when imaging through the skin. A matching liquid is usually showing dispersive effects and will consequently introduce additional complexity in the modeling. The examples we show here are mainly intended for verification of the algorithm and the FDTD model, and we therefore use an object where the use of a matching liquid is not necessary. The test object was made from sunflower oil with dielectric properties $\epsilon_r = 2.7$ and $\sigma = 0.015$ S/m at 2.3 GHz. It had the shape of a cylinder with diameter 56 mm and height 20 mm and was placed at 14 mm offset in the y-direction from the center of the antenna array. The properties of the target were constant along its height in the z-direction, and thus it is only necessary to show a cross-sectional slice of the dielectric profile. In Fig. 1(a) and (d), the original target properties have been sketched in a background of air. The measurements were made using an Agilent E8362-B PNA system equipped with the Cytec CXM/128-S-W multiplexer switch module. The multiplexer has 120 dB interchannel isolation between adjacent ports.

B. Image Reconstruction

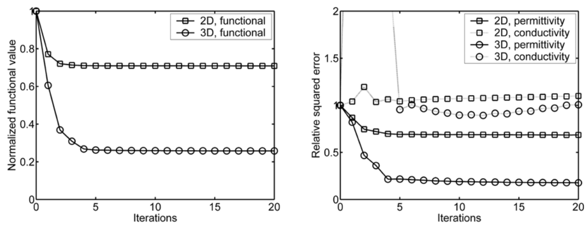
For the 3D reconstruction, we used a cubic FDTD grid with size (149 × 149 × 38) pixels with side length 2 mm and truncated by seven layers of CPML. A numerically synthesized Gaussian-envelope pulse with center frequency 2.3 GHz and full-width half-maximum bandwidth (FWHM) 2.3 GHz was used as the excitation. The height of the reconstruction domain was 20 mm, the radius was $R_{rd} = 90$ mm, and 20 iterations were performed. The resulting reconstruction of the relative permittivity and the conductivity with this pseudo-3D algorithm is shown in Fig. 1(b) and (e). The reconstructed permittivity agrees closely with the original permittivity profile, but the reconstructed conductivity deviates somewhat from the original profile. The reason being that the difference in the imaginary part of the complex dielectric permittivity between the object and background is very small compared to the difference in the real part. Reconstruction errors in the real part therefore overwhelm attempts to reconstruct the imaginary part, and consequently the accuracy is reduced.

For comparison, a reconstruction was also made from the same experimental data with the 2D reconstruction algorithm and is shown in Fig. 1(c) and (f). In the 2D algorithm, the grid was (149 × 149) pixels with size 2 mm and truncated by seven layers of CPML. The same excitation pulse as in the 3D algorithm was used. It can be seen that the reconstruction profiles for both permittivity and conductivity from the 3D algorithm are significantly more accurate compared to the profiles obtained from the 2D algorithm.

As quantitative measures of the reconstruction accuracy, we use the objective functional value, defined in (1), and the relative squared error, defined for the permittivity in (7), analogously for the conductivity. The integration of the relative squared error is made over the reconstruction domain Ω where $r < R_{rd}$.

$$\delta = \frac{\int_\Omega |\epsilon_{\text{original}} - \epsilon_{\text{reconstructed}}|^2 dS}{\int_\Omega |\epsilon_{\text{original}} - \epsilon_{\text{background}}|^2 dS}. \tag{7}$$

In Fig. 2, the relative squared error and objective functional are plotted. To enable comparison, the functional values have been normalized with respect to the initial quantities, and the improvement of the 3D algorithm over the 2D version is clearly evident. The functional values decrease with iteration number for both the 2D and 3D reconstructions and are accompanied by a corresponding decrease in the relative error for the permittivity. However, consistent with our comments above, we see that the relative squared error of the conductivity is not significantly improved. A parallel Linux cluster with 20 cores (five nodes with quad-core processors) was used to run the algorithm. The run-time for the 2D and 3D reconstructions were about



(a) (b)

Fig. 2. (a) Normalized objective functional values for the 3D/2D reconstructions versus iteration number. (b) Normalized relative squared error for the 3D/2D reconstructions of the permittivity/conductivity.

3 min and 20 h, respectively. The main reason for the increase in computation time is due to the increase of the FDTD problem in 3D compared to 2D. In Fig. 2(a), the objective functional has converged already after about five iterations, and in practice it is therefore not necessary to perform 20 iterations. With a run-time of a about 1 h per iteration, we can reduce the reconstruction time to about 5 h without significant loss of accuracy.

In [21], it was found that the most difficult object to reconstruct accurately with a 2D algorithm is a sphere. They also found that the more elongated (2D-like) the object, the better the reconstruction. They concluded that the 2D approximation is suitable for breast cancer imaging. In contrast, our results indicate that a correct accounting of 3D antenna and propagation effects results in a significant improvement in image quality over the simple 2D approach. One should also bear in mind that accurate 2D reconstructions have been published extensively in the literature and that they often use 2D-like target objects or synthetic data generated with a 2D forward solver. In conclusion, the relevance of a 2D reconstruction scheme is ultimately application-dependent.

IV. Conclusion

A 3D image reconstruction algorithm for microwave tomography with realistic modeling of the antenna system has been developed. The work was based on our earlier work with reconstructions in 2D and FDTD modeling in 3D. The antenna array was integrated into the algorithm, and successful image reconstruction of a simple test object made of sunflower oil was obtained using experimental data. Superior reconstruction accuracy was obtained using the 3D algorithm. The improvement over the 2D version can be attributed to the more accurate modeling of the antenna array response resulting in a more realistic wave propagation model. The scattering properties of the target object are also more realistically modeled in the 3D version than in 2D, where all objects are assumed to have an infinite extent in the z-direction.

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